

PSG COLLEGE OF TECHNOLOGY
DEPARTMENT OF COMPUTER APPLICATIONS
23MX26 - Java Programming Laboratory
Hands-on Worksheet 2

1. Write a program called **PrintNumberInWord** which prints "ONE", "TWO", ..., "NINE", "OTHER" if the int variable "number" is 1, 2, ..., 9, or other, respectively. Use (a) a "nested-if" statement; (b) a "switch-case-default" statement.
2. Write a program called **PrintDayInWord** which prints "Sunday", "Monday", ... "Saturday" if the int variable "dayNumber" is 0, 1, ..., 6, respectively. Otherwise, it shall print "Not a valid day". Use (a) a "nested-if" statement; (b) a "switch-case-default" statement.

Try dayNumber = 0, 1, 2, 3, 4, 5, 6, 7 and verify your results.

3. Write a program that takes three integer command-line arguments and prints equal if all three are equal, and not equal otherwise.
4. Write a program that takes the wind speed (in miles per hour) as an integer command-line argument and prints whether it qualifies as a hurricane, and if so, whether it is a Category 1, 2, 3, 4, or 5 hurricane. Below is a table of the wind speeds according to the Saffir-Simpson scale.

Category	Wind Speed (mph)
1	74 - 95
2	96 - 110
3	111 - 130
4	131 - 155
5	155 and above

5. **Gymnastics judging.** A gymnast's score is determined by a panel of 6 judges who each decide a score between 0.0 and 10.0. The final score is determined by discarding the high and low scores, and averaging the remaining 4. Write a program that takes 6 real command line inputs representing the 6 scores and prints their average, after throwing out the high and low scores.
6. **Quarterback rating.** To compare NFL quarterbacks, the NFL devised a the [quarterback rating](#) formula based on the quarterbacks number of completed passes (A), pass attempts (B), passing yards (C), touchdown passes (D), and interception (E) as follows:
 - a. Completion ratio: $W = 250/3 * ((A / B) - 0.3)$.
 - b. Yards per pass: $X = 25/6 * ((C / B) - 3)$.
 - c. Touchdown ratio: $Y = 1000/3 * (D / B)$
 - d. Interception ratio: $Z = 1250/3 * (0.095 - (E / B))$

The *quarterback rating* is computed by summing up the above four quantities, but rounding up or down each value so that it is at least 0 and at most 475/12. Write a program QuarterbackRating.java that takes five command line inputs A, B, C, D, and E, and prints the quarterback rating. Use your program to compute Steve Young's 1994 record-setting season

(112.8) in which he completed 324 of 461 passes for 3,969 yards, and threw 35 touchdowns and 10 interceptions. As of 2014, the all-time single-season record is 122.5 by Aaron Rodgers in 2011.

7. **Decimal expansion of rational numbers.** Given two integers p and q , the decimal expansion of p/q has an infinitely repeating cycle. For example, $1/33 = 0.03030303\dots$. We use the notation $0.(03)$ to denote that 03 repeats indefinitely. As another example, $8639/70000 = 0.1234(142857)$. Write a program `DecimalExpansion.java` that reads in two command line integers p and q and prints the decimal expansion of p/q using the above notation. *Hint:* use Floyd's rule.

Decimal expansion of rational numbers. Given two integers p and q , the decimal expansion of p/q has an infinitely repeating cycle. For example, $1/33 = 0.03030303\dots$. We use the notation $0.(03)$ to denote that 03 repeats indefinitely. As another example, $8639/70000 = 0.1234(142857)$. Write a C program that reads in two integers p and q and prints the decimal expansion of p/q using the above notation.

[*Hint:* Sequence of steps for $3/13$:

	Quotient	Remainder
Step-1	$3/13 = 0$	$3\%13 = 3$
Step-2	$30/13 = 2$	$30\%13 = 4$
Step-3	$40/13 = 3$	$40\%13 = 1$
Step-4	$10/13 = 0$	$10\%13 = 10$
Step-5	$100/13 = 7$	$100\%13 = 9$
Step-6	$90/13 = 6$	$90\%13 = 12$
Step-7	$120/13 = 9$	$120\%13 = 3$
Step-8	$30/13 = 2$	$30\%13 = 4$

Step-8 is the same as Step-2. Hence the decimal expansion is $0.(230769)$.]

8. **Seasons.** Write a program that takes two command line integers M and D and prints the season corresponding to month M (1 = January, 12 = December) and day D in the northern hemisphere. Use the following table.

SEASON	FROM	TO
Spring	March 21	June 20
Summer	June 21	September 22
Fall	September 23	December 21
Winter	December 21	March 20

9. **Zodiac signs.** Write a program that takes two command line integers M and D and prints the Zodiac sign corresponding to month M (1 = January, 12 = December) and day D. Use the following table

SIGN	FROM	TO
Capricorn	December 22	January 19
Aquarius	January 20	February 17
Pisces	February 18	March 19
Aries	March 20	April 19
Taurus	April 20	May 20
Gemini	May 21	June 20
Cancer	June 21	July 22
Leo	July 23	August 22
Virgo	August 23	September 22
Libra	September 23	October 22
Scorpio	October 23	November 21
Sagittarius	November 22	December 21

10. **Projectile motion.** The following equation gives the trajectory of a ballistic missile as a function of the initial angle theta and windspeed. Write a java program to print the (x, y) position of the missile at each time step t. Use trial and error to determine at what angle you should aim the missile if you hope to incinerate a target located 100 miles due east of your current location and at the same elevation. Assume the windspeed is 20 mph due east.
The equation of the trajectory is a path followed by the particle during the projectile motion.
The equation is:

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$